

Recursive methods for predictive Bayes

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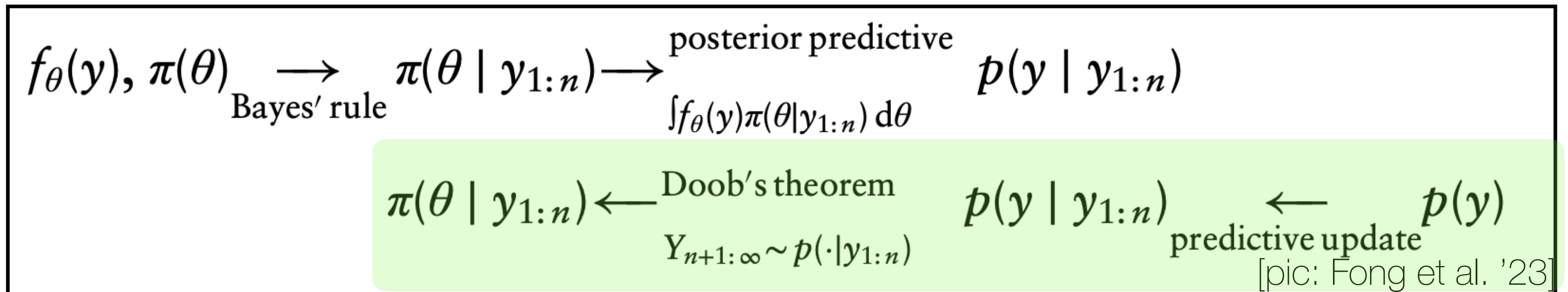
Barcelona School of Economics

Post-Bayes Seminar Series - 01/07/2025

Bayesian predictive inference

[Fortini Petrone '20, Fong et al. '23, Berti et al. '23, & others]

Let θ be a parameter of interest



- Notation: $P_n := P(Y_{n+1} \mid y_{1:n})$ and p_n the corresponding density
- $Y_{n+1:\infty} \sim P(\cdot \mid y_{1:n})$ may be tricky to elicit
- The sequence of one step ahead (P_n) is often easier to elicit
 - ▶ e.g., via a **learning rule**, i.e., a map $(P_{n-1}, y_n) \mapsto P_n$

Predictive Resampling

The mechanism that enables moving from prediction to posterior distribution

Bootstrap/Predictive Resampling [Rubin'81, Fong et al. '23]

Input:

- An estimator $\hat{\theta}_n = f(y_{1:n})$
- A learning rule: i.e., a map $(P_{n-1}, y_n) \mapsto P_n$
- Data $y_{1:n}$
- B (#bootstrap samples) and N (#synthetic samples)

Given $y_{1:n}$, compute P_n

For j in 1 to B repeat

For i in 1: N repeat

- Sample synthetic data $Y_{n+i} \sim P_{n+i-1}$
- Update the predictive $(P_{n+i-1}, y_{n+i}) \mapsto P_{n+i}$

Freedom in the choice of which predictive to use

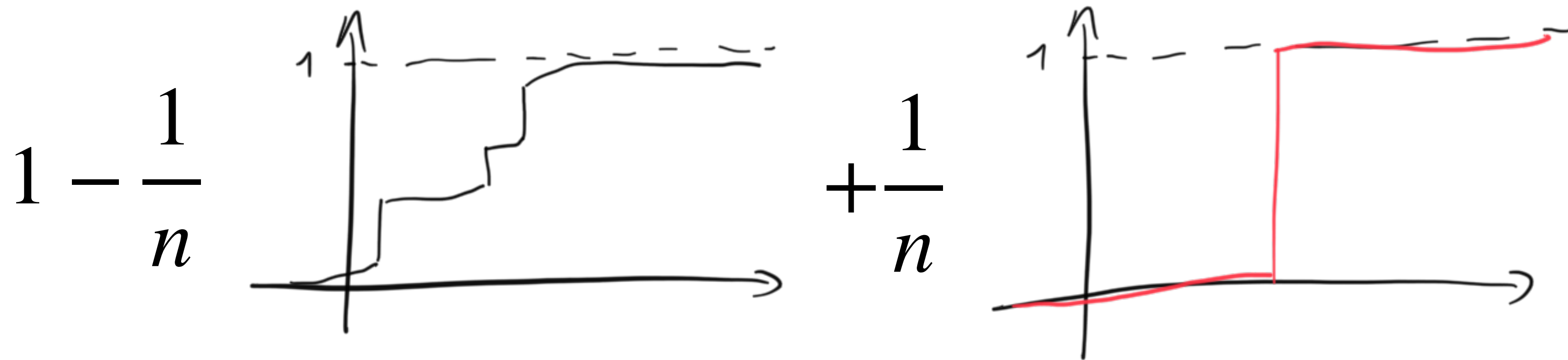
Compute $\hat{\theta}_{n+N}^{(i)} = f(y_{1:n}, y_{n+1:n+N})$

Output: $\hat{\theta}_{n+N}^{(1)}, \dots, \hat{\theta}_{n+N}^{(B)}$

Examples

Different proposals differ by how you define $(P_{n-1}, y_n) \mapsto P_n$

1.
$$P_n(\cdot) = \left(1 - \frac{1}{n}\right) P_{n-1}(\cdot) + \frac{1}{n} \delta_{Y_n}(\cdot)$$



2. ...

Here, the update is a point mass. i.e.

It's the predictive of the **Dirichlet Process**

[$\approx$ Bayesian bootstrap]

[Ferguson '73, Rubin '81]

Question 1 of the tutorial

How to come up $(P_{n-1}, y_n) \mapsto P_n$?

- A. $(P_{n-1}, y_n) \mapsto P_n$ is *given by a Bayesian model* [e.g., Dirichlet Process]
► *More interestingly*, it “mimics” the predictive distribution of a Bayesian model [e.g. Dirichlet Process Mixtures]

Today: Predictive Recursion, Gaussian Copula Algorithm

- B. $(P_{n-1}, y_n) \mapsto P_n$ is *new/almost fully new*

Today: Any-Copula Algorithm

- C. $(P_{n-1}, y_n) \mapsto P_n$ is *taken from (neighbouring) literature*

Today: Partition Estimators, Kernel methods, Parametric Bootstrap

Question 2 of the tutorial

Once we have $(P_{n-1}, y_n) \mapsto P_n$, can we use it for predictive resampling?

Two desiderata for the resulting (P_n) : (recall that $Y_n \sim P_{n-1}$)

- “Valid”: the sequence of synthetic data $(Y_N)_{N \geq n}$ should satisfy some properties
 - ▶ We will require *asymptotically exchangeable*
 - Today: - (P_n) is a **martingale** $\rightarrow (Y_n)$ is **c.i.d.** [Berti et al. '04] (Trivial for A., not B. C.)
 - (P_n) is **NOT** a martingale $\rightarrow$ **Beyond martingales** ((Y_n) is **NOT c.i.d.**)
- “Practical”: we need to be able to implement this algorithm
 - ▶ We need to sample $Y_{n+1} \sim P_n$ quickly
 - ▶ We need to update $(P_{n-1}, y_n) \mapsto P_n$ quickly
 - ▶ Other numerical properties that affect computations (grid? Order dependence? Etc.)
 - Today: not so much, except focusing on **recursive algorithm**

Constructing predictive recursion

Dirichlet Process Mixtures (DPMs)

- **Case $n = 1$:** suppose you are given a single data y_1 , the Bayes estimate of the mixing measure

$$E(G(B) | Y_1) = P(\theta_2 \in B | Y_1) = G_1(B)$$

$$= E(P(\theta_2 \in B | \theta_1) | Y_1)$$

$$= E \left(\frac{c}{c+1} G_0(B) + \frac{1}{c+1} 1_B(\theta_1) \middle| Y_1 \right)$$

$$= \frac{c}{c+1} G_0(B) + \frac{1}{c+1} P(\theta_1 \in B | Y_1)$$

Weights

$$= \frac{\int_B k(Y_i | \theta) G_0(d\theta)}{p(Y_i)}$$

Consider the mixture

$$Y_i | \theta_i \sim k(\cdot | \theta_i)$$

$$\theta_i | G \stackrel{iid}{\sim} G$$

$$G \sim DP(cG_0)$$

[Lo '84]

- It mixes **prior guess** and an “**update**”
- The update is where Bayes rule is used

- **Case $n > 2$:** usually we approximate it!

Predictive Recursion (PR) [Newton et al. '98, Newton Zhang '99]

Algo. Start with a prior guess $G_0(\cdot)$, choose a deterministic sequence (α_n) , given $y_{1:n}$

for a set B repeat

$$G_i(B) = (1 - \alpha_i)G_{i-1}(B) + \alpha_i \frac{\int_B k(y_i | \theta) G_{i-1}(d\theta)}{p_{i-1}(y_i)}$$

Output: $G_n(\cdot)$

Weights

Best guess up to $i - 1$

Update applying Bayes rule to a single obs.

Predictive Recursion (PR) [Newton et al. '98, Newton Zhang '99]

Algo. Start with a prior guess $g_0(\theta)$, choose a deterministic sequence (α_n) , given $y_{1:n}$

repeat:
$$g_i(\theta) = (1 - \alpha_i)g_{i-1}(\theta) + \alpha_i \frac{k(y_i | \theta)g_{i-1}(\theta)}{p_{i-1}(y_i)}$$

Output: $g_n(\theta)$

Weights

Best guess up to $i - 1$

Update applying Bayes rule to a single obs.

- It is exact predictive when $n = 1$
- It is not when $n > 1$
- Some numerical considerations (**order dependent**, grid, numerical integration required, ...)
- **As an estimator** It can be every accurate [Newton '02, Tokdar et al. '09, Zuanetti et al. '19, C. Walker, '18, Dixit Martin '24, ... and many more]
- [Fortini Petrone '20]: if taken as a **predictive rule**, it can be used for predictive inference

Observables? What about if we want a predictive scheme for P

Algo. Start with a prior guess $g_0(\theta)$, choose a deterministic sequence (α_n) , given

$y_{1:n}$ repeat:
$$g_i(\theta) = (1 - \alpha_i)g_{i-1}(\theta) + \alpha_i \frac{k(Y_i | \theta)g_{i-1}(d\theta)}{p_{i-1}(Y_i)}$$

Then
$$p_n(y) = \int_{\theta} k(y | \theta)g_n(\theta)d\theta$$

- It is a valid predictive rule (satisfies the martingale property), it can be used for a resampling scheme [Fortini Petrone '20]
- Lots of numerical approximations involved
- Somewhat convoluted way of getting to the sequence of (P_n)

Copula-based algorithms

Copula characterization [Hahn et al. '18]

A way to update
 $(P_{n-1}, y_n) \mapsto P_n$

Let $p_n(y) = \int f(y | \theta) \pi_n(d\theta)$ be the predictive density of Y_{n+1} given $y_{1:n}$

$$\frac{p_n(y)}{p_{n-1}(y)} = \frac{\int f(y | \theta) f(y_n | \theta) \pi_{n-1}(d\theta)}{p_{n-1}(y) p_{n-1}(y_n)}$$

Copula density
 $(P_{n-1}(y)$ is the CDF)

A. It depends only on the sample size
 B. It must converge to the independence copula (i.e. no update)

Example: Gaussian (unknown μ)

- $f(y | \theta) = N(y | \theta, 1)$ and $\pi(\theta) = N(0, \tau^{-1})$
- $p_n(y) = N(y | \mu_n, \sigma_n^2)$
- $$\frac{p_n(y)}{p_{n-1}(y)} = \frac{\int N(y | \theta, 1) N(y_n | \theta, 1) \pi_{n-1}(d\theta)}{p_{n-1}(y) p_{n-1}(y_n)}$$

Gaussian copula density
 with $\rho_n = (n + \tau)^{-1}$

Back to DPMS

$$f(y | G) = \int N(y | \theta, \sigma^2) dG(\theta) \quad \text{and} \quad G \sim DP(c G_0).$$

Case $n = 1$: suppose you are given y_1 ,

$$p_1(y) = (1 - w)p_0(y) + w \frac{\int N(y | \theta, \sigma^2) N(y_1 | \theta, \sigma^2) dG_0(\theta)}{p_0(y_1)} \quad \& \mathbf{w} \text{ is available in closed form}$$

$$\frac{p_1(y)}{p_0(y)} = (1 - w) \underbrace{1}_{\substack{\text{Independence} \\ \text{copula}}} + w \underbrace{\frac{\int N(y | \theta, \sigma^2) N(y_1 | \theta, \sigma^2) dG_0(\theta)}{p_0(y)p_0(y_1)}}_{\substack{\text{Gaussian} \\ \text{Copula}}}$$

Copula density

It is a copula mixture!

$$p_1(y) = (1 - w)p_0(y) + w c_{\rho_1}(P_0(y), P_0(y_1)) p_0(y)$$

- Similar to PR derivations
- We can use weights to drive convergence to independence copula

Gaussian copula algorithm [Hahn et al. '18]

Algo. Start with a prior guess $p_0(y)$, choose a deterministic sequence (α_n) , given $y_{1:n}$ repeat:

$$p_i(y) = (1 - \alpha_i)p_{i-1}(y) + \alpha_i p_{i-1}(y) c_\rho(P_{i-1}(y), P_{i-1}(y_i))$$

Output: $p_n(y)$

- Note, α_i drive the convergence the independence copula
- We bypass the need to compute G_n and do numerical integration
- It is exact predictive when $n = 1$
- It is not when $n > 1$
- Some numerical considerations (**order dependent**, grid, NO numerical integration required,...)
- **As an estimator** It can be every accurate [Hahn et al. '18, Fong et al. '23]
- [Fong et al. '23]: if taken as a **predictive rule**, it can be used for predictive inference

Same as PR

Nothing special about location-DPM

Let $f(y | G) = \int N(y | \theta, \sigma^2) dG(\theta, \sigma^2)$ and $G \sim DP(c G_0)$. Given y_1 ,

$$\frac{p_1(y)}{p_0(y)} = (1 - w)1 + w c_{\rho, \nu}(P_0(y), P_1(y)) \quad \text{Student t copula density}$$

Algo. Start with a prior guess $p_0(y)$, choose a deterministic sequence (α_n) , given $y_{1:n}$ repeat:

$$p_i(y) = (1 - \alpha_i)p_{i-1}(y) + \alpha_i p_{i-1}(y) c_{\rho, \nu}(P_{i-1}(y), P_{i-1}(y_i))$$

Output: $p_n(y)$

- If taken as a sequence of predictives, it defines a martingale
- Same reasoning can be extended to more complex models
 - ▶ Multivariate/Regression [Fong et al. '23]
 - ▶ DPM of Linear Regression [C. Walker '25a]

Intermediate “lessons”

- Starting from a Bayesian model, it is fairly easy to get a martingale
- Practical algorithms
- Can we avoid taking a Bayesian model as starting point?

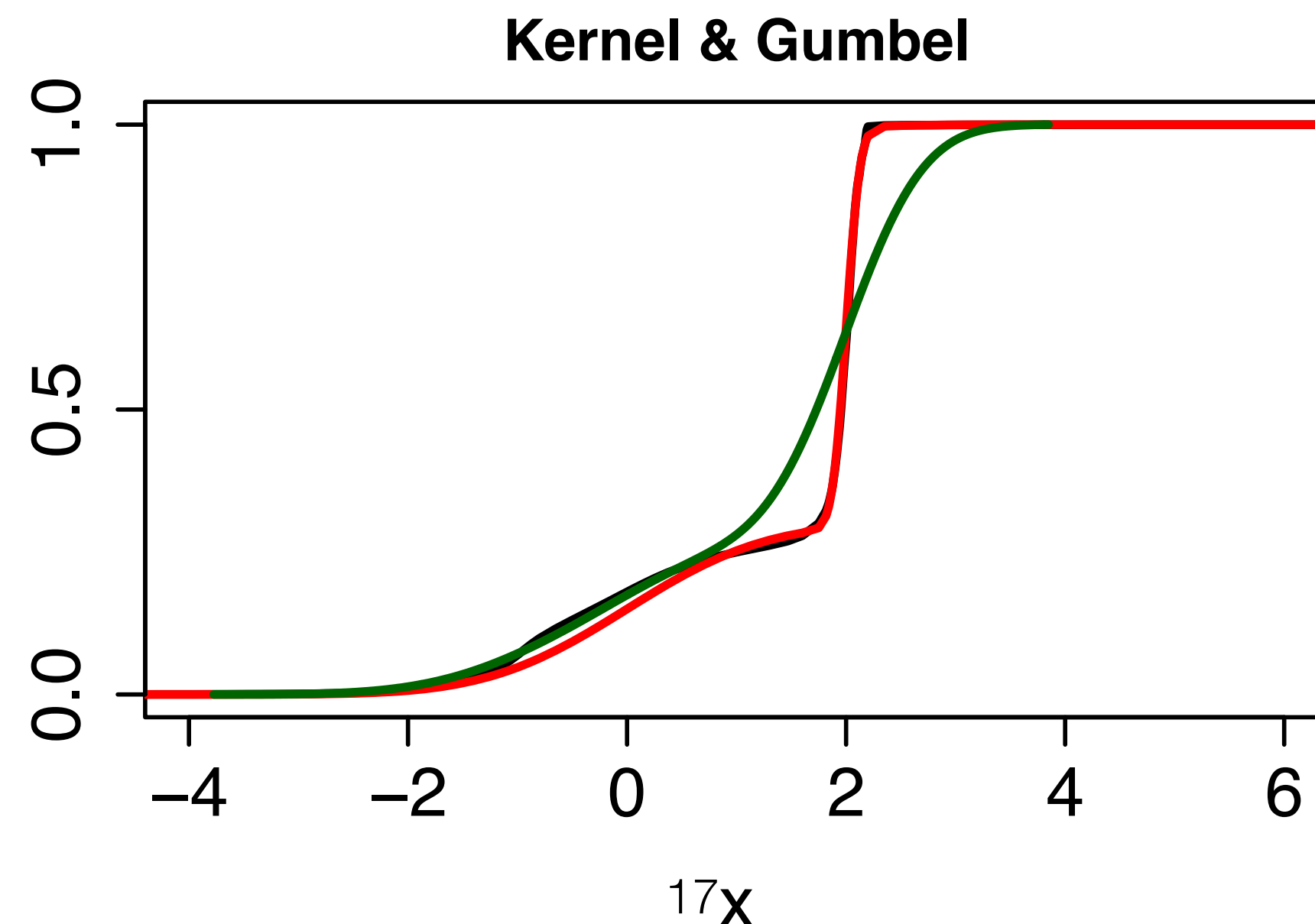
Any-copula

Algo. Start with a prior guess $p_0(y)$, choose a deterministic sequence (α_n) , given $y_{1:n}$ repeat: $p_i(y) = (1 - \alpha_i)p_{i-1}(y) + \alpha_i p_{i-1}(y) c_\theta(P_{i-1}(y), P_{i-1}(y_i))$

Output: $p_n(y)$

- We did not start from a (known) Bayesian model
- We chose a copula c_θ and use the weights as before
- Example: Gumbel copula [C. Walker '25b]

Black: truth
Green: kernel density estimator
(Silverman's rule)
Red: Algo. With Gumbel Copula



Can we use it for
predictive
resampling?

Validity: **Martingale condition**

$$\mathbb{E}(P_{n+1}(B) | Y_{1:n}) =$$

$$\mathbb{E} \left(\int_B p_n(y) c_\theta(P_n(y), P_n(Y_{n+1})) dy \middle| Y_{1:n} \right) = \iint_B p_n(y) p_n(y_{n+1}) \frac{p(y, y_{n+1})}{p_n(y) p_n(y_{n+1})} dy dy_{n+1} = P_n(B)$$

- The resulting sequence is (Y_n) is c.i.d.
- The connection between copulas and c.i.d. sequences is deeper [Bissiri Walker '25]

- We can sample from copula-based predictive algorithms [Fong et al. '23]
- It is clearly very fast

Copulas define a “valid”
 $(P_{n-1}, y_n) \mapsto P_n$

$Y_{n+1} \sim P_n$ is
“practical”

Beyond Bayesian models

(Recursive) Partition estimator [Tukey '47,'61, Colomb '77,...]

Consider

- $(\mathcal{P}_n)$: sequence of nested dyadic partitions of $[0,1]$, with
 $\mathcal{P}_0 = \{A_0 = [0,1]\}$, $\mathcal{P}_1 = \{A_{1,0} = [0,1/2), A_{1,1} = [1/2,1]\}$...
- $A_n(x)$: the cell containing x

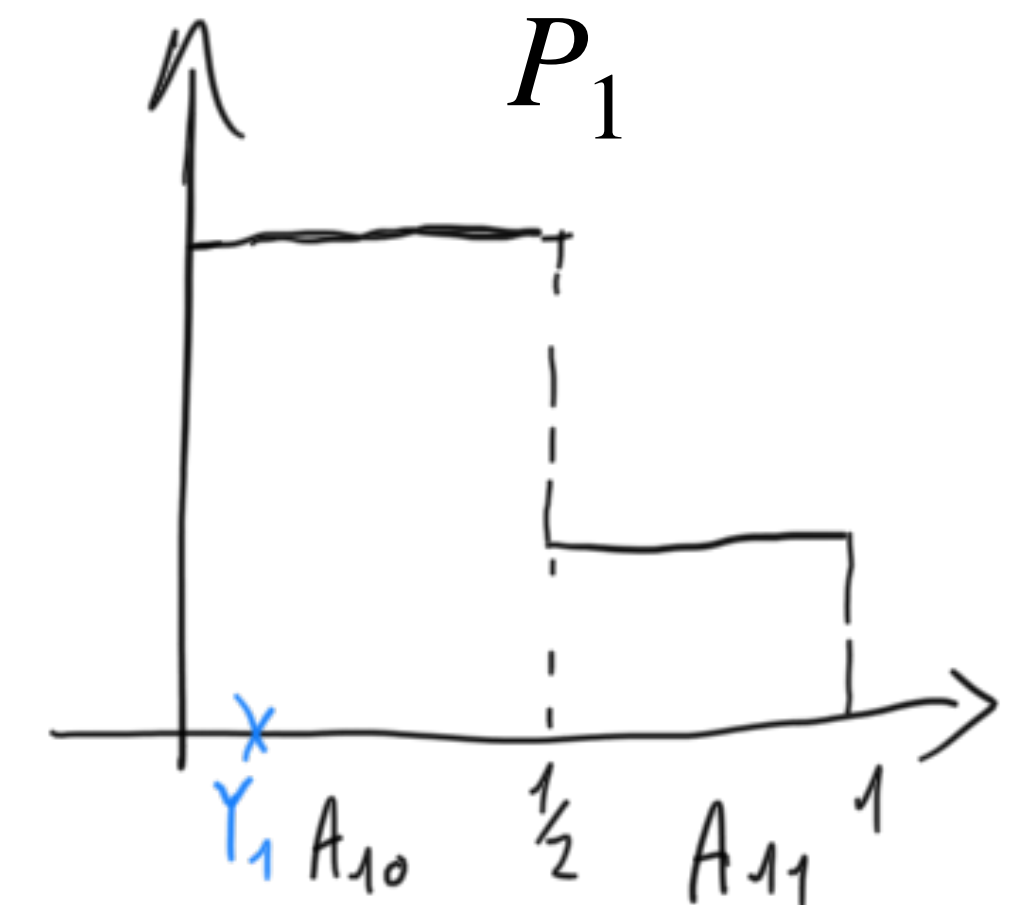
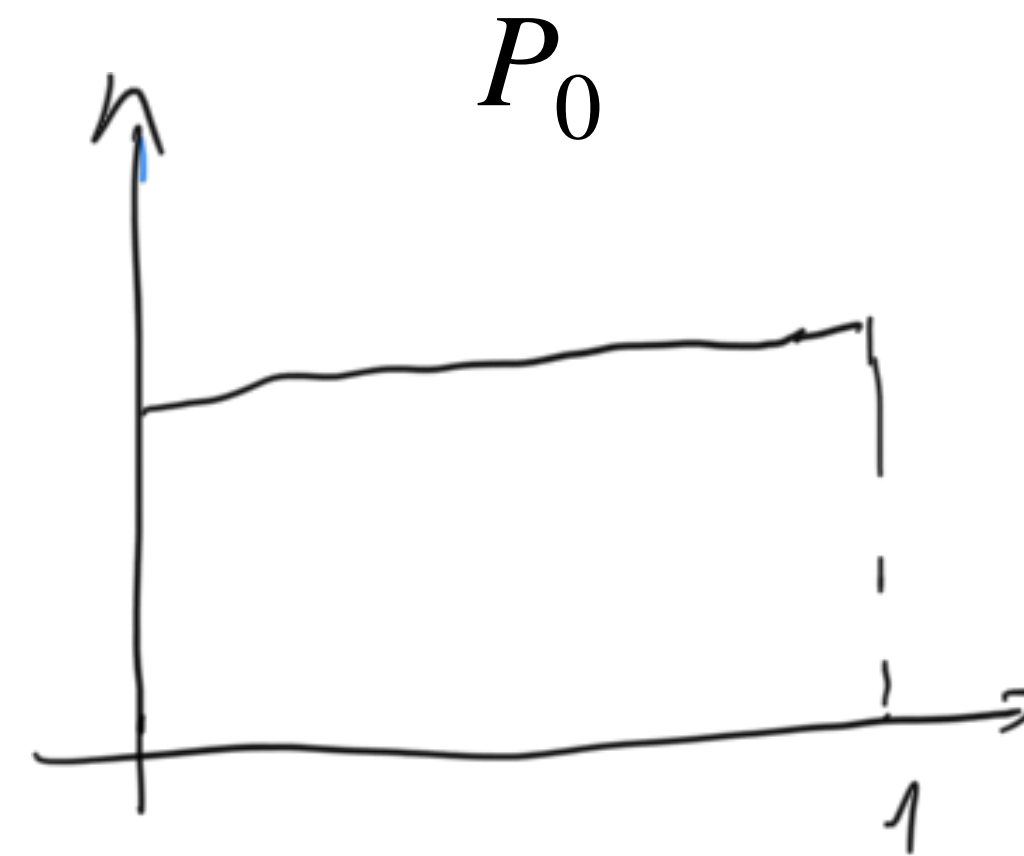
Recursive predictive scheme

$$Y_1 \sim \text{Unif}(A_0) := P_0$$

$$Y_2 | Y_1 \sim \frac{1}{2} \text{Unif}(A_0) + \frac{1}{2} \text{Unif}(A_1(Y_1)) := P_1$$

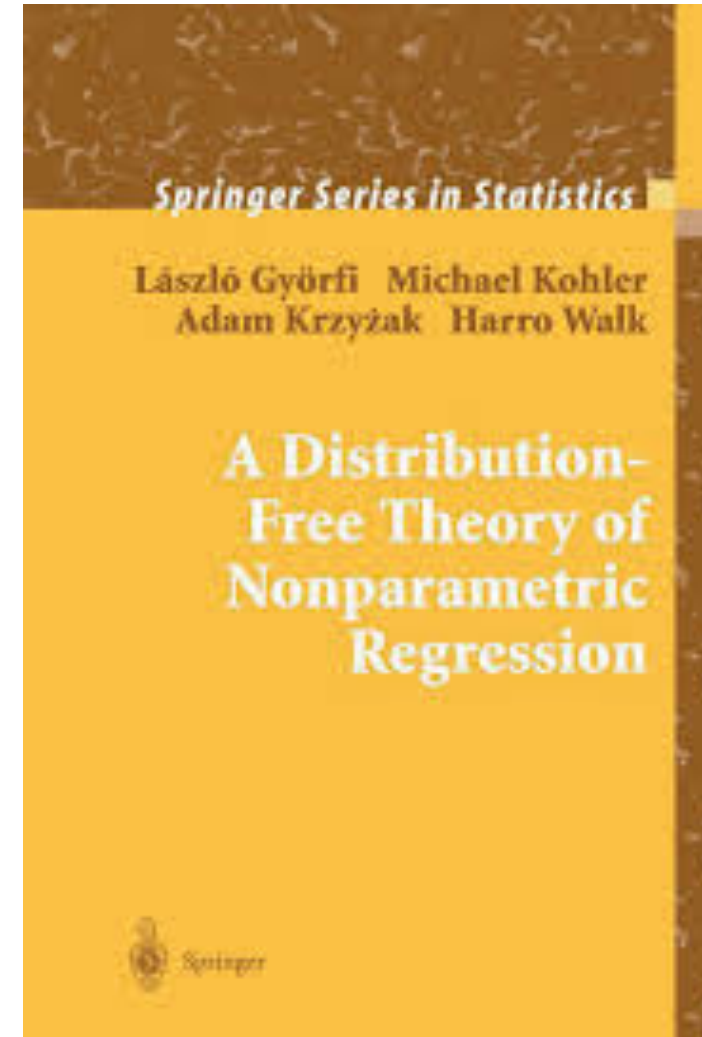
...

$$Y_{n+1} | Y_{1:n} \sim \left(1 - \frac{1}{n}\right) P_n + \frac{1}{n} \text{Unif}(A_n(Y_n)) := P_{n+1}$$



[Gyorfi et al.
'12, Ch. 4,25]

- Update $(P_{n-1}, y_n) \mapsto P_n$ is “valid”: (P_n) is martingale.
- It can be used for nonparametric regression etc.
- Not sure how “practical” it is. (Definitely order dependent etc.)

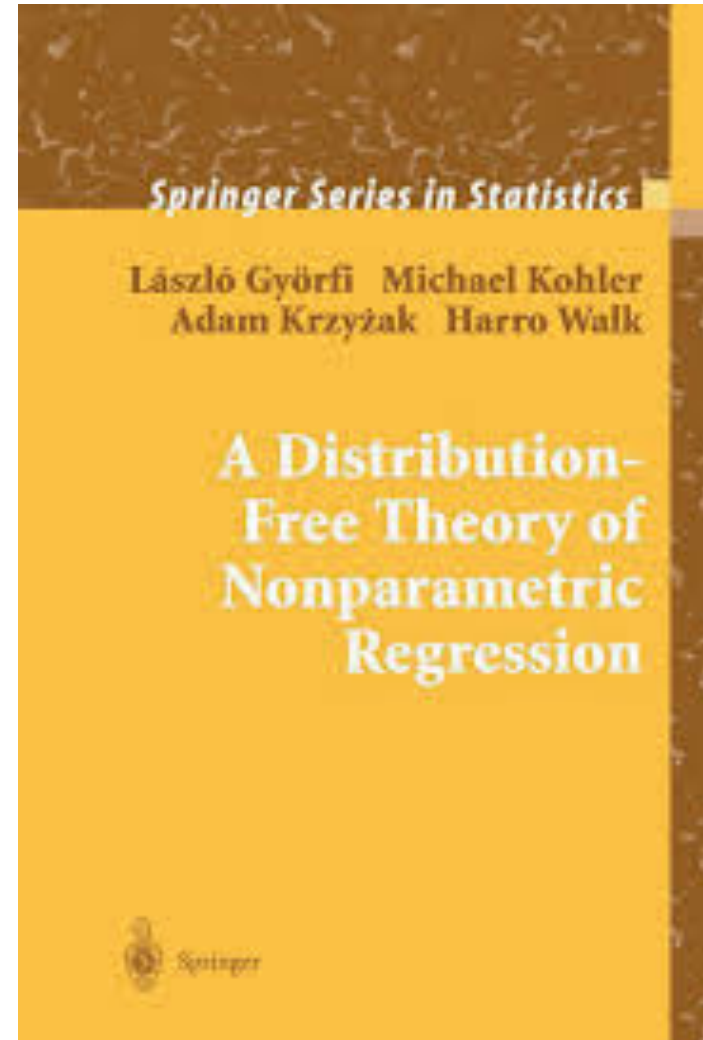


(Recursive) Kernel estimator [Parzen '62, Rosenblatt'56, Akaike '54,...]

Consider

- Kernel $K : \mathbb{R} \mapsto \mathbb{R}^+$ such that $\int K(u)du = 1$
- Bandwidth (h_n) with $h_n \rightarrow 0$

[Gyorfi et al.
'12, Ch. 5, 25]



Recursive predictive scheme

$$Y_1 \sim K := P_0$$

- $Y_{n+1} | Y_{1:n} \sim P_n^c$ with $p_n^c(x) = \frac{1}{nh_n} \sum_{i=1}^n K\left(\frac{x - Y_i}{h_n}\right)$ (classical kernel estimator)
- $Y_{n+1} | Y_{1:n} \sim P_n^r$ with $p_n^r(x) = \left(1 - \frac{1}{n}\right) P_{n-1}^r + \frac{1}{nh_n} K\left(\frac{x - Y_n}{h_n}\right)$ (recursive kernel estimator)

- (P_n^c) is a martingale if and only if Kernel is Laplace with a specific scale [Thm1 West '91]
- (P_n^r) is not a martingale [Battiston & C. '25]

Beyond Martingales

Parameter updates

Several proposals [Holmes Walker '23, Garelli et al. '24, Fong Yiu '24, Fortini Petrone '25...] moved the recursive update to a parametric family. With appropriate initialization, **repeat**

$$Y_{n+1} | \hat{\theta}_n \sim P_{\hat{\theta}_n}$$
$$\hat{\theta}_n = \hat{\theta}_{n-1} + \eta_n f(\hat{\theta}_{n-1}, Y_n, \dots)$$

Example [Holmes Walker '23]: Gaussian

- $Y_{n+1} \sim N(\hat{\theta}_n, 1)$
- $\hat{\theta}_n = \bar{Y}_n$ (sample mean)

Is this “valid” update?

Validity: **Martingale condition**

$$\mathbb{E}[P_{n+1}(B) | Y_{1:n}] = \mathbb{E}[N(B | \bar{Y}_{n+1}, 1) | Y_{1:n}]$$

$$= N\left(B \mid \bar{Y}_n, 1 + \frac{1}{(n+1)^2}\right)$$

Not a martingale!

What other options we have?

For a general sequence (P_n) , if all we are trying to establish is that the resulting (Y_n) sequence is **asymptotic exchangeability**, we need to show [Aldous '85, Lemma 8.2]

$$\mathbb{P}(\omega \in \Omega : P_n(\cdot, \omega) \xrightarrow{w} \tilde{F}(\cdot, \omega)) = 1$$

[Aldous '85]

- If (P_n) not a martingale, the c.i.d. property is lost
- This makes establishing the above somewhat non-trivial, as we don't even have this conditional identical distributed property

Convergence of parameters/moments

- **$(\hat{\theta}_n)$ is martingale:** Prove parameter convergence using martingales property, and then one is practically done
 - ▶ Parametric Bayesian Bootstrap [Holmes Walker '23, Fong Yiu '24]
 - ▶ Logistic regression (Sandra's talk) [Fortini Petrone '25]
- **Convergence of sample mean and sample variance for non i.i.d. random variables** appropriately constructed
 - ▶ Predictive distributions driven by 1st and 2nd sample moments [Garelli et al. '25]

Both approaches cover the Gaussian example and much more general examples

Larger classes of a.s. weakly convergent r.m.s.

- (P_n) belongs to a class of random measures a.s. weakly convergent

▸ [Battiston & C. '25] Informally, if (P_n) satisfies

$$|P(X_{n+2} \in B | Y_{1:n}) - P(X_{n+1} \in B | Y_{1:n})| \leq \xi_n \text{ and } \sum \xi_n < \infty \text{ a.s.}$$

- The martingale property is acquired asymptotically
- It covers the Gaussian example
- It also covers recursive kernel estimators (not-order dependent in the data, no grid required, easy to sample)
- Talk on YouTube - Post Bayes Workshop (same channel as the seminar series)

Discussion

Discussion

- We have reviewed several possible constructions of algorithms that can be used for these **bootstrap-type schemes**
 - ▶ They can come mimicking $n = 1$ update of existing Bayesian models (Predictive Recursion, Gaussian Copula,...)
 - ▶ They can be new (Any-copula, Predictive sequence driven by moments, SGD driven predictives,...)
 - ▶ They can be borrowed from other literature (Recursive Partition estimators,...)
- We have discussed ways of checking their suitability for these schemes
 - ▶ “Valid” update (Martingale conditions, ...)
 - ▶ “Practical” (Being able to sample from, fast update,...)

Themes not touched (Maybe simply because there is not enough out there)

- Which scheme to use?
 - ▶ For specific applications, it will depend on “*suitability to the available data*” [Garelli et al. '24]
 - ▶ Still, maybe something can be done to understand the properties. I like a “*forward ($n + 1 : \infty$)-backward ($1 : n$)*” view in [Fong Yiu '24]
- More practical ways of establishing validity
 - ▶ All approaches require some form of mathematical tractability

Thanks

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References (In the order at which they appeared)

- Fortini, S., S. Petrone, et al. (2020). Quasi-Bayes properties of a procedure for sequential learning in mixture models. *Journal of the Royal Statistical Society Series B* 82(4), 1087–1114.
- Fong, E., C. Holmes, and S. G. Walker (2023). Martingale posterior distributions. *Journal of the Royal Statistical Society Series B* 85(5), 1357–1391.
- Berti, P., Dreassi, E., Pratelli, L. & Rigo, P. (2021). A class of models for Bayesian predictive inference. *Bernoulli*, 27, 702–726.
- Rubin, D. B. (1981). The Bayesian bootstrap. *The Annals of Statistics*, 9(1), 130–134.
- Newton, M. A., Zhang, Y., & Régnier, D. (1998). Nonparametric Bayes methods using predictive updating. In J. M. Bernardo et al. (Eds.), *Bayesian Statistics 6* (pp. 651–660). Oxford University Press.
- Newton, M. A., & Zhang, Y. (1999). A recursive algorithm for nonparametric Bayes classification. *Biometrika*, 86(2), 239–249.
- Newton, M. A. (2002). On a nonparametric recursive estimator of the mixing distribution. *Sankhyā: The Indian Journal of Statistics, Series A*, 64, 306–322.
- Tokdar, S. T., Martin, R., & Ghosh, J. K. (2009). Consistency of a recursive estimate of mixing distributions. *The Annals of Statistics*, 37(5B), 2502–2522.
- Zuanetti, D.A., Müller, P., Zhu, Y., Yang, S., Ji, Y.: Bayesian nonparametric clustering for large data sets. *Statistics and Computing* 29, 203–215 (2019)
- Cappello. Walker (2018). A Bayesian motivated Laplace inversion for multivariate probability distributions. *Methodology and Computing in Applied Probability*
- Dixit, V., & Martin, R. (2023). Revisiting consistency of a recursive estimator of mixing distributions. *Electronic Journal of Statistics*, 17(1), 1007-1042.

References (In the order at which they appeared)

- Hahn, P.R., Martin, R., Walker, S.G. (2018) On recursive predictive distributions. *Journal of the American Statistical Association* 113, 1085–1093
- Cappello, L, Walker, S.G. (2025a). Recursive nonparametric predictive of a regression model.
- Cappello, L, Walker, S.G. (2025b). Recursive estimation of probability distributions
- Bissiri, P. G., & Walker, S. G. (2025). Bayesian analysis with conditionally identically distributed sequences. *Electronic Journal of Statistics*, 19(1), 1609-1632.
- Tukey, J. W. (1947). Non-parametric estimation II. Comparison of distributions. *Annals of Mathematical Statistics*, 18(4), 501–507.
Tukey, J. W. (1961). Curves as parameters, and touch estimation. In *Proceedings of the Fourth Berkeley Symposium on Mathematical Statistics and Probability*.
- Györfi, L., Kohler, M., Krzyżak, A., & Walk, H. (2012). A distribution-free theory of nonparametric regression. Springer Science & Business Media.
- Parzen, E. (1962). On estimation of a probability density function and mode. *The Annals of Mathematical Statistics*, 33(3), 1065–1076.
- Rosenblatt, M. (1956). Remarks on some nonparametric estimates of a density function. *The Annals of Mathematical Statistics*, 27(3), 832–837.
- Akaike, H. (1954). An approximation to the density function. *Annals of the Institute of Statistical Mathematics*, 6, 127–132.
- West, M. (1991). Kernel density estimation and marginalization consistency. *Biometrika* 78(2), 421–425.

References (In the order at which they appeared)

- Battiston, M., Cappello, L. (2025) New (and old) predictive schemes with a.c.i.d. sequences
- Holmes, C. C. and S. G. Walker (2023). Statistical inference with exchangeability and martingales. *Philosophical Transactions of the Royal Society A* 381(2247), 20220143.
- Garelli, S., F. Leisen, L. Pratelli, and P. Rigo (2024). Asymptotics of predictive distributions driven by sample means and variances. *arXiv preprint arXiv:2403.16828*.
- Fong, E. and A. Yiu (2024). Asymptotics for parametric martingale posteriors. *arXiv preprint arXiv:2410.17692*.
- Fortini, S. and S. Petrone (2025). Exchangeability, prediction and predictive modeling in bayesian statistics. *Statistical Science* 40(1), 40–67.